

SOFTWARE ENGINEERING

01

Overview of IT Industry

Understand how the IT world works — roles, tools, and the journey from idea to software.

DURATION	LEVEL	MODULE	UPDATED
5 Sessions	Beginner	Module 1	May 2026

IN THIS MODULE

01	Welcome to the IT World
02	How Software Works
03	Internet & Networking Basics
04	Software Development Lifecycle (SDLC)
05	Git & Version Control

SESSION

01

60 MIN · IT INDUSTRY OVERVIEW

Welcome to the IT World

◆ Learning Objectives

- Identify the key roles in an IT company and describe what each does day-to-day
- Distinguish between product and service companies using real Indian examples
- Recognise the collaboration tools — GitHub, Slack, Jira — used by real software teams
- Navigate a live open-source GitHub project and identify different contributor roles
- Explore emerging tech job roles shaping India's IT sector in 2025

01 Who Works in IT?

India is home to over 5 million IT professionals — the largest tech workforce in the world. Companies like TCS, Infosys, and Wipro operate out of Bengaluru, Hyderabad, Pune, and Ahmedabad, hiring thousands of graduates every year. Gujarat-based firms like Infibeam Avenues (a fintech company from Ahmedabad) are also building global products with local talent.

CONCEPT — IT Roles at a Glance

Every software product is built by a team of specialists. A Developer writes the code, a Tester finds what breaks before users do, a Designer makes it easy and beautiful to use, a Data Analyst reads the numbers to make decisions, and an AI Engineer trains models that power smart features. When you use the Zomato app — all five of these roles contributed to what you see on screen today.

Role	What They Do	Common Tools
Developer	Write and maintain code for features	VS Code, GitHub, IntelliJ
Tester / QA	Find and log bugs before users do	Selenium, Postman, Jira
UI/UX Designer	Design screens, flows, and user experience	Figma, Adobe XD
Data Analyst	Turn raw data into business decisions	Python, Excel, Tableau
AI / ML Engineer	Build and train intelligent models	TensorFlow, PyTorch, Python

02 Product vs Service Companies

Not all IT companies work the same way. Some build their own product and sell it to the world — these are product companies. Others build software on contract for other businesses — these are service companies. Understanding this difference is one of the first career decisions you will make.

Product Company

Builds and owns its own product. Revenue comes from users, subscriptions, or licences. Examples: Zoho (Chennai), Google, Freshworks. You go deep on one product — your work directly shapes what millions of customers use every day.

Service Company

Builds software on project contracts for client businesses. Revenue comes from billing hours or project fees. Examples: TCS, Infosys, Wipro. You work across many industries and clients — exposure is broad and hiring volumes are high.

PRO TIP — Which One Should You Target?

Service companies hire in large volumes, making them a great first entry point. Product companies typically pay more and move faster but are more selective. Most TOPS alumni start at service companies and transition to product roles within 2–3 years as their portfolio and skills grow.

CASE STUDY — Zoho Corporation

Context — Zoho is an Indian software product company headquartered in Chennai. It builds cloud-based business tools — CRM, Finance, HR, Marketing automation — and sells them to over 100 million users across 180 countries.

Challenge — In the early 2000s, Zoho competed against giants like Salesforce and Microsoft with no external funding and very limited brand recognition outside India.

What they did — Instead of outsourcing or chasing investors, Zoho built entirely in-house — their own engineering teams, their own university (Zoho Schools of Learning that recruits from small-town India), and their own cloud infrastructure. They focused on making world-class software at a fraction of competitors' prices.

Result — Today Zoho is a \$1 billion+ revenue company with zero external investment, and one of India's most respected product companies. They hire thousands of engineers directly from tier-2 and tier-3 cities across India including Gujarat.

Takeaway — You don't need Silicon Valley funding or a pedigree college to build a world-class product — Zoho proved it from India.

Source: Zoho Corporation official website & The Economic Times, 2023

03 How IT Teams Collaborate

Software is never built by one person. A team of developers, testers, designers, and managers needs a shared system to stay organised, communicate, and track work. The three most important tools are: GitHub for code, Slack for communication, and Jira for task management.

Tool	Used For	Analogy
GitHub	Store code, track all changes, collaborate on features and fixes	Google Drive for code — with a full history of who changed what and when
Slack	Team messages organised by channel, file sharing, integrations	WhatsApp for offices — but organised by project and department
Jira	Task boards, sprint planning, bug tracking with priorities	To-do list for the entire team — with deadlines, owners, and status tracking

TRY THIS LIVE

Open github.com/sindresorhus/awesome in your browser right now. This is a real, very popular open-source project. Click 'Contributors' on the right side — see how many people from different countries (including India) have built this together. Click on any contributor's profile to see what they work on.

04 Emerging Tech Roles in India

The IT industry is shifting fast. Traditional roles like manual tester or data entry operator are shrinking; new roles like DevOps Engineer, Prompt Engineer, Cloud Architect, and Cybersecurity Analyst are growing rapidly. NASSCOM projects India will need 1 million AI-skilled professionals by 2026 — and companies are already struggling to find qualified candidates.

IT PROFESSIONALS IN INDIA

5.4M+

Source: NASSCOM IT Industry Report, 2024

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Software Engineering by IIT Bombay — Lecture 1: Introduction to Software Engineering-Challenges

Link — <https://nptel.ac.in/courses/106101061>

Always link the official channel — beware fan-made or re-upload copies.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a career counsellor for IT students in India. List the top 5 emerging tech jobs in India in 2025, explain what each role does in simple Hinglish, name one real Indian company actively hiring for each role, and say what one skill to learn first for each. Keep the total response under 150 words.

WHERE TO GO NEXT

You now know who works in IT, how companies are structured, and what tools real software teams use every day.

Explore deeper — Visit [linkedin.com/jobs](https://www.linkedin.com/jobs) and search 'Software Engineer Ahmedabad' — open 3 job descriptions and note which tools and skills appear most often.

Try on your own — Ask ChatGPT: 'What is the difference between a product and a service IT company in India? Give me 5 real examples of each.' Compare the list with what you now know from this session.

Connect the dots — Next session we go inside the software itself — what it is made of, how it is structured in layers, and how your phone talks to a server thousands of kilometres away.

Going further — Read Chapter 1 of 'The Pragmatic Programmer' by Hunt and Thomas — it covers the professional mindset every serious developer needs, and it is available as a preview on Google Books.

SESSION

02

60 MIN · IT INDUSTRY OVERVIEW

How Software Works

◆ Learning Objectives

- Define what a software application is and name its three core components
- Explain the 3-layer architecture — UI, Business Logic, and Database — with examples
- Compare desktop, web, and mobile applications using real Indian products
- Trace a complete client-server request from a user tap to the database and back
- Draw a basic data flow diagram for a familiar app using Draw.io or Lucidchart

01 What is a Software Application?

A software application is a program that helps a user accomplish a specific task — book a train, transfer money, or watch a video. Every app you use daily, from PhonePe to Google Maps, is a collection of code, business rules, and stored data working together invisibly in the background.

CONCEPT — The Three Core Components

Every software application has exactly three parts: the User Interface (what you see and click), Business Logic (the rules that decide what happens), and the Database (where all data is stored permanently). When you search 'vada pav near me' on Zomato, the UI sends your request, the business logic finds and ranks matching restaurants, and the database returns their menus, photos, and prices — all in under a second.

02 The 3-Layer Architecture

Almost every software system — from an IRCTC booking portal to a bank app — is deliberately organised in three separate layers. Keeping layers separate means that changes in one layer don't break the others, and different teams can work independently without conflicts. This principle is called separation of concerns.

Layer	Also Called	What It Does	Example in Swiggy
UI / Presentation	Front-end	What the user sees and interacts with	Restaurant listing screen, search bar, cart page
Business Logic	Back-end	The rules, calculations, and decisions	Filter by rating, calculate ETA, apply discount codes
Database	Data / Storage layer	Stores and retrieves all persistent data	Menus, user addresses, order history, restaurant ratings

PRO TIP — Why Layers Save Teams

When Swiggy redesigns its app (UI changes), the database does not need to change. When they update their delivery fee calculation (business logic), the app screens stay exactly the same. Separate layers let 100+ engineers work simultaneously without breaking each other's work — this is why every professional software system uses this pattern.

03 Types of Applications

Software can be delivered in three different environments. Knowing the type of application you are building — or using — shapes every technology decision you will make as a developer.

Desktop App

Installed directly on a computer. Works offline.
Examples: Tally ERP (used by most shops and CAS across Gujarat), MS Office, VLC. Fast and powerful — but updates must be installed manually and the app is tied to that specific machine.

Web App

Runs inside a browser. No installation needed.
Examples: Gmail, GST Portal (gstn.org.in), Google Docs. Works on any device with internet — always up to date, but requires connectivity and can be slower than a native desktop app.

CONCEPT — Mobile Apps

Mobile apps are installed on a phone (Android or iOS). Unlike websites, they can access phone hardware directly — camera, GPS, microphone, accelerometer, and push notifications. Examples: BHIM UPI, PhonePe, Jio Cinema. Many companies like HDFC Bank run both a web app AND a mobile app that both connect to the exact same back-end and database behind the scenes — the layers are shared.

04 Client-Server Communication

Every time you open Instagram or check your IRCTC ticket, a conversation happens between your device (the client) and a remote computer in a data centre thousands of kilometres away (the server). This is called the client-server model and it is the single most important concept powering the modern internet.

CONCEPT — The Request-Response Cycle

Step 1 — You tap Refresh on Instagram (your phone sends a request to Instagram's server in Mumbai or Singapore). Step 2 — The server receives the request and queries its database for your friends' latest posts. Step 3 — The database returns matching records to the server. Step 4 — The server packages the data and sends a response back to your phone. Step 5 — Your app renders the new posts on screen. This entire round trip typically completes in under 300 milliseconds.

TRY THIS LIVE

Open Chrome -> press F12 (or right-click -> Inspect) -> click the Network tab -> now open swiggy.com or instagram.com. Watch the requests appear in real time as the page loads. Click any row — you can see the exact URL called, what data was sent, and what came back. That is live client-server communication happening right in front of you.

CASE STUDY — Flipkart's 3-Layer Architecture

Context — Flipkart, India's largest e-commerce platform (now owned by Walmart), serves 400 million+ registered users, including millions of customers in Gujarat cities like Surat, Vadodara, and Ahmedabad.

Challenge — During Big Billion Days sale events, Flipkart receives 100x its normal traffic in seconds. A flat, single-layer system would crash instantly under this load.

What they did — Flipkart built a strict 3-layer architecture: a lightweight mobile and web front-end, a microservices-based business logic layer (each service handles exactly one job — pricing, inventory, orders, or payments), and a distributed database that replicates data across multiple data centres across India so no single failure brings down the platform.

Result — Flipkart successfully handled over 11 million orders in a single day during its 2023 Big Billion Days event with no significant downtime.

Takeaway — The same 3-layer architecture taught in this session is what powers India's biggest platforms — understanding it deeply is what separates junior from senior developers.

Source: Flipkart Engineering Blog, 2023 & Economic Times, Big Billion Days coverage

“Software is not magic — it is layers of simple decisions, stacked carefully on top of each other.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Software Engineering by IIT Bombay — Lecture 3 & 4: Overview of Phases (covers software architecture and design layers)

Link — <https://nptel.ac.in/courses/106101061>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a software architect. Explain the 3-layer architecture — UI, Business Logic, Database — using the example of booking a train ticket on IRCTC in simple Hinglish. Walk me through step-by-step what happens from when I click 'Search Trains' to when I see the results on screen. Keep it under 150 words.

WHERE TO GO NEXT

You can now explain how any app you use works — from the screen you touch to the database that stores your data.

Explore deeper — Open diagrams.net (Draw.io — free in browser) and draw the 3-layer architecture for any one app of your choice. Even a simple WhatsApp message flow will work.

Try on your own — Pick any app on your phone — PhonePe, Paytm, or BookMyShow. Write down what the UI does, what decisions the back-end makes, and what the database stores. Compare your answer with a classmate.

Connect the dots — But how does your device even reach that remote server? Next session we go deeper — into the internet itself: IP addresses, DNS, protocols, and how your data physically travels across networks.

Going further — Read 'The Architecture of Open Source Applications' — free at aosabook.org — Chapter 1 shows how real production systems are structured using the same principles taught today.

SESSION

03

60 MIN · IT INDUSTRY OVERVIEW

Internet & Networking Basics

◆ Learning Objectives

- Explain how computers communicate over the internet using packets and protocols
- Identify common protocols — HTTP, HTTPS, FTP, SMTP — and when each is used
- Describe how DNS translates a domain name into an IP address step by step
- Distinguish between IPv4 and IPv6 and explain why the shift is happening in India
- Read HTTP status codes (200, 404, 500) and interpret what each one means
- Use browser DevTools Network tab and the nslookup command to inspect live traffic

01 How Computers Communicate

The internet is not a single wire — it is a massive, global network of computers, routers, and undersea cables all connected together. When you send a WhatsApp message from Ahmedabad to Mumbai, your message does not travel as one piece. It is broken into small chunks called packets, each sent along the fastest available route, and then perfectly reassembled at the other end.

CONCEPT — Packets and the Internet

Think of sending a large parcel from Ahmedabad to Delhi via courier. Instead of one heavy box, you break it into 10 smaller boxes and send each via the fastest available route — some via Jaipur, some via Mumbai. They may arrive in different order, but the recipient reassembles them using numbered labels. That is exactly what the internet does with your data — every video, email, and message is broken into numbered packets and reassembled perfectly at the destination.

02 Protocols — The Language of the Internet

For two computers to communicate, they must agree on a shared set of rules. These agreed rules are called protocols. Different protocols handle different jobs — HTTP transfers web pages, SMTP handles email, and FTP transfers files. Without protocols, computers cannot understand each other.

Protocol	Full Form	What It Does	Indian Example
HTTP	HyperText Transfer Protocol	Transfers web pages from server to browser	Loading any public website like irctc.co.in
HTTPS	HTTP Secure	Same as HTTP but encrypted — data cannot be read in transit	Logging in to SBI NetBanking or making a GPay payment

Protocol	Full Form	What It Does	Indian Example
FTP	File Transfer Protocol	Transfers files between two computers over a network	Uploading website files to a hosting server
SMTP	Simple Mail Transfer Protocol	Sends email from one mail server to another	When you send email from Gmail or Rediffmail

HTTP

Unencrypted. Data travels in plain text — anyone on the same Wi-Fi network could read it. Safe only for public information pages with no login or payment. Browsers now show a 'Not Secure' warning on HTTP sites.

HTTPS

Encrypted using TLS/SSL. Data is scrambled in transit — even if intercepted, it cannot be read. Mandatory for any login, payment, or personal data page. Look for the padlock icon and 'https://' in the URL before entering any sensitive data.

WARNING — Never Enter Passwords on HTTP Sites

If a website URL starts with 'http://' (no 's') and asks for your password or card number — do not enter it. Without HTTPS, your credentials travel in plain text and can be read by anyone on the same Wi-Fi. Always check for the padlock icon before entering any sensitive information.

03 Domains, DNS & IP Addresses

Every device on the internet has a unique numerical address called an IP address — similar to how every house in Ahmedabad has a unique postal PIN code. But remembering numbers like 142.250.77.46 is hard, so we use domain names like google.com. DNS (Domain Name System) is the service that translates human-readable domain names into numeric IP addresses that computers actually use.

CONCEPT — DNS: The Internet's Phone Book

When you type 'tops-int.com' in your browser, your computer doesn't know where that server is. It asks a DNS server: 'What IP address does tops-int.com map to?' The DNS server replies with the correct IP address. Your browser then connects directly to that IP address and loads the page. This DNS lookup happens in milliseconds, every time you open any website — silently, invisibly, billions of times a day worldwide.

◆ DNS Hierarchy — From Root to Subdomain

Level	Example	What It Represents
Root	.	The invisible top of all DNS — managed globally by ICANN

Level	Example	What It Represents
Top-Level Domain (TLD)	.com .in .org .edu	The last part of a domain name — .in is India's country code
Domain	tops-int google irctc	The unique name registered by a company or individual
Subdomain	mail.google.com api.tops-int.com	A prefix that organises different services under one domain

```
nslookup google.com
```

 Look up the IP address of google.com — run in Command Prompt (Windows) or Terminal (Mac/Linux)

```
nslookup tops-int.com
```

 Find the IP address of the TOPS Technologies website

```
nslookup irctc.co.in
```

 Find the IP address of the IRCTC train booking portal

◆ IPv4 vs IPv6 — Why Addresses Are Running Out

IPv4

Format: 4 numbers separated by dots. Example: 192.168.1.1. Supports approximately 4.3 billion unique addresses. We have nearly run out — India alone has over 800 million internet users competing for available IPv4 addresses.

IPv6

Format: 8 groups of 4 hex digits. Example: 2001:0db8:85a3::8a2e:370:7334. Supports 340 trillion trillion unique addresses — essentially unlimited. Jio and BSNL have already begun nationwide IPv6 rollout across India.

04 HTTP Lifecycle & Status Codes

Every time your browser loads a page, it follows an HTTP lifecycle: send a request, receive a response. That response always includes a three-digit status code — the server's way of saying 'everything worked,' 'page not found,' or 'something crashed on our end.' Every developer must know these cold.

Code	Meaning	When You See It
200 OK	Request succeeded — everything is fine	Every normal page load (you just never notice it)
301 Moved Permanently	The page has moved to a new URL forever	When a website changes its domain or restructures URLs
400 Bad Request	The browser sent a malformed or invalid request	Submitting a form with missing required fields

Code	Meaning	When You See It
404 Not Found	The page or file does not exist on this server	Clicking a broken link or typing a wrong URL
500 Internal Server Error	Something went wrong on the server side	When IRCTC or a bank website crashes during peak hours

TRY THIS LIVE

Open Chrome -> press F12 -> go to the Network tab -> visit swiggy.com. Look at the Status column — you will see 200 codes for successful loads and possibly 301 redirects. Now type any URL that doesn't exist on a live website — you will get a 404 response. You are now reading live HTTP status codes directly from a real server.

CASE STUDY — Jio and India's Internet Revolution

Context — Before September 2016, mobile internet in India cost Rs 250–Rs 350 per GB. Most people in smaller Gujarat cities like Rajkot, Bhavnagar, and Anand could not afford to browse freely — internet was a luxury.

Challenge — Reliance Jio needed to build a network and software platform — JioTV, MyJio, JioSaavn — that could handle hundreds of millions of first-time internet users simultaneously, many on weak 4G connections in rural areas.

What they did — Jio built a 100% 4G VoLTE network with aggressive data compression, HTTP response caching, and adaptive streaming protocols. These protocols automatically lowered video quality on weak signals to keep apps responsive — the same HTTP and networking principles taught in this session.

Result — Within 6 months, Jio added 100 million subscribers — the fastest telecom ramp in world history. Data costs fell 95%. India climbed from 155th in global internet speed rankings to the top 15 within 5 years.

Takeaway — The protocols and networking principles you learned today are exactly what Jio's engineers used to connect 450 million Indians to the internet.

Source: TRAI Telecom Subscription Report 2022 & Reliance Industries Annual Report 2022–23

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Computer Networks and Internet Protocol by IIT Kharagpur — Week 1: Introduction to Computer Networks & Week 3: Application Layer Services (HTTP, FTP, DNS)

Link — <https://nptel.ac.in/courses/106105183>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a networking teacher. Explain how DNS works step-by-step in simple Hinglish — from the moment I

type 'swiggy.com' in my browser to when the page starts loading. Use the analogy of asking someone for directions in Ahmedabad. Keep the total response under 150 words.

WHERE TO GO NEXT

You can now explain what the padlock in your browser means, how nslookup works, and what a 404 error actually is — these are real developer skills.

Explore deeper — Visit whatismyipaddress.com to see your current public IP address. Then run nslookup on 3 websites you use daily — see exactly which IP address each domain resolves to.

Try on your own — Open any website in Incognito Mode with the Network tab open. Count how many separate HTTP requests are made just to load the homepage — most sites make 50–100 requests. See if you can identify which are 200s and which are 301s.

Connect the dots — Now that you understand how software is structured AND how it communicates across networks, the next session brings it all together — the Software Development Lifecycle (SDLC) that turns an idea into a shipped product.

Going further — Read the Cloudflare Learning Center article 'What is DNS?' at developers.cloudflare.com/learning/dns/what-is-dns — it has excellent interactive diagrams and goes deeper into recursive DNS resolution.

SESSION

04

60 MIN · IT INDUSTRY OVERVIEW

Software Development Lifecycle (SDLC)

◆ Learning Objectives

- List and explain each of the 6 SDLC phases in the correct sequence
- Distinguish between the Waterfall and Agile development methodologies
- Describe the three Scrum roles and what each one is responsible for
- Relate each SDLC phase to a real-world app feature release with an Indian example
- Read a Jira-style sprint board and identify where work sits at each stage

01 What is the SDLC?

The Software Development Lifecycle (SDLC) is the structured process every professional software team follows to plan, build, and deliver software reliably. Without it, projects miss deadlines, exceed budgets, and ship with critical bugs — this is the most common failure in student and early-startup projects. With it, a team of 50 engineers can ship a feature every two weeks without chaos.

CONCEPT — Why SDLC Matters

Imagine building a 'live bus tracking' feature for the GSRTC booking portal. Without SDLC, developers might start coding before requirements are finalised, testers might only check the feature one day before launch, and the deployment might happen on a busy Monday without a rollback plan. The SDLC prevents every one of these failures by defining what happens, in what order, and who owns each step.

02 The 6 SDLC Phases

The standard SDLC has six phases. Each phase has a clear input, defined activities, and an output that the next phase depends on. A feature does not move forward until the current phase is reviewed and signed off. This discipline is what separates professional software teams from hobby coders.

Phase	What Happens	Output / Deliverable	Who Leads
1. Requirements	Gather what the software must do — from business and users	Requirements document or user stories	Business Analyst / Product Manager
2. Design	Plan the architecture, UI layout, and database structure	System design document, wireframes, ER diagram	Senior Developer / Architect / Designer

Phase	What Happens	Output / Deliverable	Who Leads
3. Development	Write the actual code based on the design specifications	Working code in a version-controlled repository	Developers
4. Testing	Find and fix defects before the software reaches users	Test reports, bug-fix log, sign-off document	QA Engineers / Testers
5. Deployment	Release the tested software to real users in production	Live software on production servers	DevOps / Release Manager
6. Maintenance	Monitor, fix issues, and ship improvements post-launch	Patches, updates, and new sprint cycles	Entire team on a continuous basis

TRY THIS LIVE

Open github.com/microsoft/vscode/releases — this is VS Code's real release history. Read the latest release notes. Try to identify which SDLC phase each item came from: a new feature (Design + Development), a bug fix (Testing -> Maintenance), or a performance improvement (Requirements -> Deployment). Real SDLC output, right in front of you.

03 Agile vs Waterfall

There are two dominant ways to apply the SDLC in practice. Waterfall runs the six phases in a straight line — you fully complete each phase before starting the next. Agile repeats the SDLC in short cycles called sprints, delivering small working pieces of software every 1–4 weeks and adjusting direction based on real feedback.

Waterfall

All requirements defined upfront. Phases run in strict sequence — going back is expensive. Works best for fixed-scope projects with clear requirements: government portals, defence systems, large banking core infrastructure. Long gap between project start and first working product.

Agile

Requirements evolve sprint by sprint based on feedback. Small, working software is delivered every 1–4 weeks. Works best for consumer apps, startups, and any product where market feedback shapes direction. Used by Swiggy, Zepto, Flipkart, and most modern software teams.

COMPANIES USING AGILE GLOBALLY

71%

Source: PMI Pulse of the Profession Report, 2023

04 Scrum — Agile in Practice

Scrum is the most widely used Agile framework. It organises a team into three specific roles, runs work in time-boxed sprints of 1–2 weeks, and uses a visual task board to track progress daily. Almost every IT company you join — from an Ahmedabad startup to a Bengaluru MNC — uses some version of Scrum.

Scrum Role	Responsibility	Analogy
Product Owner (PO)	Decides what to build next and in what priority — owns and maintains the product backlog	The client or vision-holder — says 'build this first'
Scrum Master	Removes blockers, runs ceremonies, protects the team from distractions	A coach — keeps the team moving at full speed
Development Team	Designers, developers, and testers who build the product together	The builders — they commit to delivering in each sprint

CONCEPT — The Jira Sprint Board

A Jira sprint board has four columns: Backlog (work planned but not started), To Do (assigned to the current sprint), In Progress (actively being worked on now), and Done (complete and reviewed). Every task — called a user story or ticket — moves left to right as work progresses. Any manager can look at the board at any moment and instantly see the status of the entire sprint without calling a single meeting.

CASE STUDY — How Zepto Ships Features Using Agile

Context — Zepto, India's fast-growing 10-minute grocery delivery startup, operates in Ahmedabad, Mumbai, Bengaluru, and other major cities. It competes directly on speed — both in delivery and in software feature launches.

Challenge — Zepto had to build and iterate its app extremely fast — adding features like scheduled delivery, wallet payments, and live order tracking within weeks of competitors launching similar features.

What they did — Zepto adopted Agile with 2-week sprints. Each sprint has a single clear goal — one feature or one improvement. The Product Owner prioritises the backlog based on customer data and feedback. The dev team builds, QA tests, and the feature ships at the end of every sprint.

Result — Zepto grew from zero to Rs 2,500 crore in annual revenue in under 3 years, shipping new app features nearly every two weeks. By 2023 it reached a valuation of \$1.4 billion.

Takeaway — Agile did not make Zepto fast — it gave the team a discipline that made reliable, repeated shipping possible at high speed.

Source: Zepto Engineering Blog & Economic Times, 2023

“Working software delivered regularly beats a perfect plan delivered late.”

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Software Project Management by Prof. Rajib Mall, IIT Kharagpur — Week 1: Introduction to Software Project Management (covers SDLC phases, planning, and estimation)

Link — <https://nptel.ac.in/courses/106105218>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a Scrum Master at a Mumbai tech startup. Explain the Scrum framework to a new team member in simple Hinglish — cover the 3 Scrum roles, what a sprint is, how the Jira board works, and what happens in a sprint review. Use a real Indian product like Zepto or PhonePe as the example. Keep it under 150 words.

WHERE TO GO NEXT

You now understand how professional software teams plan, structure, and ship products using SDLC, Waterfall, and Agile/Scrum.

Explore deeper — Visit atlassian.com/agile/scrum — Atlassian (the makers of Jira) have a free, well-illustrated guide to every Scrum concept covered today.

Try on your own — Take any feature you use daily — say, Zomato's 'Reorder last meal' button — and map out what each of the 6 SDLC phases might have looked like when that feature was built.

Connect the dots — The final session of Module 1 completes the picture — how code is tracked and shared across a team using Git and GitHub, the version control tool every professional developer uses every day.

Going further — Read 'The Scrum Guide' at scrumguides.org — it is the official Scrum reference, only 13 pages, and it is free. Every serious developer should read it once.

SESSION

05

60 MIN · IT INDUSTRY OVERVIEW

Git & Version Control

◆ Learning Objectives

- Explain why version control is essential for any professional software project
- Define the six core Git concepts: Repository, Commit, Branch, Merge, Push, and Pull
- Initialise a local Git repository and make a first commit from the terminal
- Push a local project to GitHub and view the commit history online
- Describe how Pull Requests and Forks enable team collaboration on GitHub

01 Why Version Control?

Imagine you are building a website for a Surat textile business. You spend 3 days adding new features — and then realise they broke a form that was working before. Without version control, you have no way to go back. You would manually undo everything from memory and hope for the best. Version control eliminates this problem entirely: it keeps a complete, timestamped record of every change ever made to your project.

Without Version Control

Files named: project_final.js, project_FINAL_v2.js, project_USE_THIS_ONE_FINAL.js. No record of what changed or why. One wrong save and the previous version is gone forever. Team members email zip files to each other. Nobody knows whose version is correct.

With Git

One clean project folder. Every saved change has a message, timestamp, and author. Jump back to any version in seconds. The full team works from one shared repository — no zip files, no confusion, no overwriting each other's work.

CONCEPT — What Version Control Does

Think of Git like the version history in Google Docs — but for your entire codebase. Every time you save a version (called a commit), Git records exactly what changed, who changed it, and a message explaining why. If a bug appears in production, a developer can review every change made in the last 24 hours and pinpoint exactly which line introduced the problem — in minutes, not days.

02 Core Git Concepts

Git has six foundational concepts. Every Git command you will ever run is built on these six ideas. Master the concepts first — the commands will then make logical sense rather than feeling like magic words you have to memorise.

Concept

What It Is

Simple Analogy

Concept	What It Is	Simple Analogy
Repository (Repo)	The tracked project folder — contains all files and their full history	A project archive that remembers every version it has ever been
Commit	A saved snapshot of your project at one specific point in time	Pressing Save in Photoshop — but with a label, timestamp, and author
Branch	An independent copy of the code where you work without affecting others	A parallel draft — edits here don't touch the main document
Merge	Combining the changes from one branch back into the main branch	Accepting tracked changes in a Word document back into the master copy
Push	Uploading your local commits to the remote server on GitHub	Syncing your local changes up to the shared cloud copy
Pull	Downloading the latest changes from GitHub to your local machine	Syncing the team's latest updates down to your computer

PRO TIP — Write Meaningful Commit Messages

Bad commit messages: 'fixed stuff', 'changes', 'asdfgh'. Good commit messages: 'Fix login button not responding on Safari mobile' or 'Add WhatsApp OTP field to registration form'. A good commit message tells any future reader — including yourself 3 months later — exactly what changed and why. Every professional team has a commit message standard; build this habit from your very first commit.

03 Your First Git Commands

The core Git workflow is the same every single day for every developer at every company in the world: initialise a repo, make changes, stage the changes, commit them with a message, and push to GitHub. These steps become automatic muscle memory within your first week of real use.

```
git init           Initialise a new Git repository in the current folder
git status         Show which files have been changed but not yet
                  committed
git add .          Stage all changed files to be included in the next
                  commit
git commit -m "Add homepage layout for TOPS project"
                  Save a named snapshot of all staged files with a descriptive message
git remote add origin https://github.com/your-username/your-repo.git
                  Link your local repository to a remote GitHub repository
```

```
git push -u origin main
```

Upload your committed changes to GitHub for the first time

```
git pull origin main
```

Download and merge the latest team changes from GitHub to your machine

```
git log --oneline
```

See a compact summary of every previous commit in this project

```
git checkout -b feature/new-login-page
```

Create and switch to a new branch for a separate feature

TRY THIS LIVE

Open your terminal. Create a new folder anywhere on your computer and navigate into it. Run 'git init', then create a text file (any content), then run 'git add .' followed by 'git commit -m "My first commit"'. Finally, run 'git log --oneline' — you will see your commit in the project history. You have just created a version-controlled project from scratch.

04 GitHub — Collaboration at Scale

Git manages your local project history. GitHub is the cloud platform where teams store, share, and collaborate on their Git repositories. Your GitHub profile is your public portfolio — every commit you push, every project you publish, and every Pull Request you open is visible to recruiters and hiring managers across India and worldwide.

GitHub Feature	What It Does	When You Use It
Fork	Create your own copy of someone else's public repository in your account	When you want to contribute to an open-source project
Pull Request (PR)	Propose that your branch's changes be merged into the main branch	After finishing a feature — you request teammates to review your code
Issues	Report bugs or request features on any project	When you find a problem or have a suggestion for any project
GitHub Actions	Automatically run tests and deployments on every code push	CI/CD — code automatically tested and deployed without manual steps

CONCEPT — The Pull Request Workflow

In every professional team, no developer pushes directly to the main branch. The workflow is: 1) Create a new branch for your feature. 2) Commit your work on that branch. 3) Push the branch to GitHub and open a Pull Request. 4) A teammate reviews your code and leaves comments. 5) After approval, the branch is merged into main and deployed. This process catches bugs before they ever reach users — it is used by every company from Infosys to Google.

CASE STUDY — India's Open Source Rise on GitHub

Context — India is consistently among the top 3 countries contributing to open-source projects on GitHub, with developers from Ahmedabad, Pune, Bengaluru, and Hyderabad contributing to global software used by millions of people.

Challenge — Many Indian students and junior developers have solid coding skills but no visible, verifiable portfolio — making it harder to get interview calls at product companies that screen GitHub profiles before scheduling interviews.

What they did — Thousands of Indian developers began contributing to open-source projects on GitHub — starting small with documentation fixes and typo corrections, progressing to bug fixes, and eventually building new features. Each accepted Pull Request is permanently linked to their GitHub profile.

Result — Indian developers have become core maintainers of major global projects including VS Code extensions, React ecosystem libraries, and widely-used Python packages. A strong GitHub profile has directly helped hundreds of Indian developers land roles at Google, Microsoft, and Atlassian.

Takeaway — Every public commit you push on GitHub is a permanent, verifiable proof of your skills — it is the most powerful portfolio tool available to a developer.

Source: GitHub Octoverse Report 2023 & LinkedIn Talent Insights India

PRO TIP — Use AI to Debug Git Errors

When you see a Git error — 'merge conflict', 'rejected push', 'detached HEAD state' — copy the exact error message from your terminal and paste it into GitHub Copilot Chat, ChatGPT, or Claude. Ask: 'What does this Git error mean and how do I fix it step by step?' You will get a precise fix in seconds. Looking up errors is a professional skill, not a weakness — every developer at every level does this every day.

WATCH — VIDEO REFERENCE (ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Software Engineering by IIT Bombay — lectures covering Software Configuration Management and version control in the software lifecycle

Link — <https://nptel.ac.in/courses/106101061>

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a senior developer at a Pune IT company. Explain Git branching and the Pull Request workflow to a fresher joining their first job in simple Hinglish. Use the analogy of a group college project — where 5 students are all editing the same Word document — to show how Git solves the problem of overwriting each other's work. Keep it under 150 words.

WHERE TO GO NEXT

Module 1 is complete. You can now explain the IT industry, how software is built in layers, how data travels across the internet, how teams manage work with SDLC and Scrum, and how code is tracked with Git.

Explore deeper — Create a free account at github.com if you don't have one. Push any project — even a folder with one HTML file — and share the public repository link with your trainer.

Try on your own — Fork any public repository on GitHub. Make one small change — fix a spelling mistake in the README file — and open a Pull Request back to the original repo. That is your first open-source contribution.

Connect the dots — Module 2 starts the hands-on building. You will create real web pages with HTML and CSS — and use Git to version-control every project from session one.

Going further — Complete the free 'Introduction to Git and GitHub' course on Coursera by Google (coursera.org) — it takes 4–5 hours and gives you a verifiable certificate to add to your LinkedIn profile.

REVIEW



INTERVIEW PREP · MODULE 1

Interview Questions

◆ SE – Overview of IT Industry

Rehearse each answer out loud, aiming for about 60 seconds. The hint is only a nudge — try answering first, then compare with the model answer below.

1

What is the difference between a product-based and a service-based IT company? Give an example of each.

Think about whether the company builds its own products or delivers solutions for clients.

MODEL ANSWER

A product-based company builds and sells its own software product, such as Google selling its search engine and ads platform. A service-based company takes projects from clients and delivers custom solutions, like TCS or Infosys building software for external businesses. Product companies focus on continuous improvement of one platform, while service companies manage many different client projects simultaneously.

2

Name three common career roles in the IT industry and briefly describe what each one does.

MODEL ANSWER

A developer writes and maintains the code that powers software applications. A tester verifies that the software behaves correctly by designing and running test cases. A data analyst examines datasets to draw insights and support business decisions. Other roles include UI/UX designer, DevOps engineer, and AI/ML engineer.

3

How do teams use tools like GitHub, Slack, and Jira to collaborate on software projects?

Focus on what each tool helps the team achieve.

MODEL ANSWER

GitHub is used for version control — developers push code, review pull requests, and track code history. Slack is the communication hub where teams share updates, ask questions, and receive automated notifications from other tools. Jira is used for project management, where tasks (called stories or tickets) are created, assigned, and tracked through stages like To Do, In Progress, and Done. Together these tools keep the whole team aligned on code, communication, and delivery progress.

4

What is a software application, and what are its main components?

Think about what layers or parts make up most apps.

MODEL ANSWER

A software application is a program designed to help users perform specific tasks, such as messaging, banking, or shopping. Its main components are the User Interface (UI) that users interact with, the Business Logic layer that processes rules and decisions, and the Database that stores and retrieves data. These three layers work together to handle every user action from clicking a button to saving a record.

5

Explain the three main layers in typical software architecture: UI, Business Logic, and Database.

MODEL ANSWER

The UI (User Interface) layer is what the user sees and interacts with — buttons, forms, and screens. The Business Logic layer is the brain of the application; it processes inputs, applies rules, and decides what to do next. The Database layer stores all persistent data such as user profiles and transactions. A request flows from UI to Business Logic to Database and back, making all three layers essential for a working application.

6

Describe, step by step, what happens when you refresh your feed in a social media app.

MODEL ANSWER

When you pull to refresh, the app sends an HTTP request to the backend server asking for the latest posts. The server's business logic queries the database for new content relevant to your account. The database returns the records, the server formats them into a response, and sends it back to the app. The UI then re-renders to display the new posts on screen. This full round-trip typically takes under a second on a good network.

7

Briefly explain what each of these protocols is used for: HTTP, HTTPS, FTP, SMTP.

MODEL ANSWER

HTTP (HyperText Transfer Protocol) is used to transfer web pages between a browser and a server. HTTPS is the secure version of HTTP — it encrypts the data using SSL/TLS so that nobody can intercept it. FTP (File Transfer Protocol) is used to upload or download files between computers. SMTP (Simple Mail Transfer Protocol) is used to send emails from one server to another.

8

What is a domain name, and how does DNS help users reach a website?

Think about how human-friendly names are linked to IP addresses.

MODEL ANSWER

A domain name is a human-readable address like `www.tops-technologies.com` that is easier to remember than a numeric IP address. DNS (Domain Name System) acts like a phone book for the internet — it translates the domain name into the actual IP address of the server. When you type a URL, your browser queries a DNS server, receives the IP address, and then connects to the web server at that address. Without DNS, users would need to memorize numeric IPs for every site.

9

What command can you use to find the IP address of a website like `google.com`, and what does it show you?

MODEL ANSWER

The command is ``nslookup google.com`` run in the terminal or command prompt. It queries the DNS system and returns the IP address (or addresses) mapped to that domain, along with the DNS server that answered the query. This is useful for diagnosing DNS problems, checking which server a domain resolves to, or verifying DNS propagation after a domain change.

10**What is an IP address, and how is IPv4 different from IPv6?***Consider the format and length of each type.***MODEL ANSWER**

An IP address is a unique numerical label assigned to every device on a network, used to identify and communicate with it. IPv4 uses a 32-bit format written as four numbers separated by dots (e.g., 192.168.1.1), supporting about 4.3 billion unique addresses. IPv6 uses a 128-bit format written in hexadecimal groups separated by colons, supporting virtually unlimited addresses. IPv6 was introduced because the world is running out of IPv4 addresses due to the explosion of internet-connected devices.

11**When you type a URL in your browser and hit Enter, what is the HTTP request and response lifecycle? Mention common status codes.****MODEL ANSWER**

The browser first resolves the domain via DNS to get the server's IP address, then opens a TCP connection and sends an HTTP GET request. The server processes the request and returns an HTTP response with a status code and body. A 200 status means success, 404 means the requested resource was not found, and 500 means an internal server error occurred. The browser then renders the returned HTML, CSS, and JavaScript to display the page.

12**List and briefly describe the main phases of the Software Development Lifecycle (SDLC).****MODEL ANSWER**

The main SDLC phases are: Requirement Gathering (understanding what the client needs), Design (planning the system architecture and UI), Development (writing the actual code), Testing (finding and fixing bugs), Deployment (releasing to production), and Maintenance (updating and supporting the live system). Each phase feeds into the next, ensuring the final product is reliable and meets user needs.

13**What is the difference between the Agile and Waterfall models of software development?***Think about how work is structured and delivered in each.***MODEL ANSWER**

Waterfall is a linear sequential model — each phase must be fully completed before the next begins, making it harder to accommodate changes midway. Agile is iterative — work is broken into short sprints (typically 2 weeks) and the product is delivered in increments, allowing continuous feedback and adjustments. Waterfall suits projects with fixed, well-understood requirements, while Agile is better for projects where requirements evolve. Most modern teams prefer Agile for its flexibility and faster delivery cycles.

14**In Scrum, what are the roles of Product Owner, Scrum Master, and Development Team?****MODEL ANSWER**

The Product Owner defines and prioritizes the product backlog — the list of features and requirements — and represents the client or business. The Scrum Master facilitates the Scrum process, removes blockers for the team, and ensures Scrum practices are followed. The Development Team does the actual work — designing, coding, and testing — and is self-organizing. All three roles collaborate during sprint planning, daily stand-ups, reviews, and retrospectives.

15

Why is version control important in software development projects?**MODEL ANSWER**

Version control tracks every change made to the codebase over time, so you can see who changed what and when. It allows multiple developers to work on the same project simultaneously without overwriting each other's work, using branches. If a new change breaks something, you can roll back to a previous working version. Tools like Git are the industry standard, and platforms like GitHub add collaboration features like pull requests and code reviews.

16

Explain these Git concepts: Repository, Commit, Branch, Merge, Push, and Pull.**MODEL ANSWER**

A repository (repo) is a folder that Git tracks — it contains all project files and their history. A commit is a saved snapshot of your changes with a message describing what was done. A branch is a separate line of development that lets you work on a feature without affecting the main code. Merge combines changes from one branch into another. Push uploads your local commits to a remote repository like GitHub, and Pull downloads the latest changes from the remote repo to your local machine.

17

How does a Pull Request work on GitHub, and why is it important for collaboration?**MODEL ANSWER**

A Pull Request (PR) is a proposal to merge code from one branch into another, typically from a feature branch into the main branch. It lets other team members review the code, leave comments, request changes, and approve before the merge happens. PRs act as a quality gate — they catch bugs, enforce coding standards, and ensure everyone on the team is aware of changes. This makes collaboration structured and traceable.

18

Describe the steps to initialize a Git repo, add files, commit, and push to GitHub.**MODEL ANSWER**

First, run ``git init`` inside your project folder to initialize a local repository. Use ``git add .`` to stage all files for the next commit, then ``git commit -m 'Initial commit'`` to save the snapshot. On GitHub, create a new empty repository, then link it with ``git remote add origin <URL>``. Finally, run ``git push -u origin main`` to upload your code to GitHub. After this first push, ``git push`` alone is enough for future commits.

19

Imagine you are working with a team on a project using GitHub and Jira. How would you use these tools together to track progress and manage code changes?**MODEL ANSWER**

Each Jira ticket represents a task or user story. A developer picks a ticket, creates a Git branch named after the ticket ID (e.g., TOPS-42-login-page), and works on it. When done, they push the branch and open a Pull Request on GitHub, linking it to the Jira ticket. After code review and merge, the Jira ticket is moved to Done. This workflow ensures every line of code is tied to a requirement, keeping both code history and project tracking in sync.

20

If a website is not loading and you suspect a DNS issue, what steps would you take to diagnose it?

Consider checking the domain's resolution and the commands you might use.

MODEL ANSWER

Start by running ``nslookup <domain>`` or ``ping <domain>`` in the terminal to see if the domain resolves to an IP address. If it doesn't resolve, the DNS record may be missing or misconfigured. Check the domain's DNS settings in your registrar or DNS provider to verify the A record points to the correct server IP. You can also use online tools like dnschecker.org to confirm DNS propagation across global servers. If DNS is resolving correctly but the site still fails, the issue may be on the server side.